Numerical Analysis I

Quiz 10A Name:_____

Consider the nonlinear model

$$y(t;c) = \frac{c}{1+ct}$$

and the two data points

$$(t_1, y_1) = (0, 2), \qquad (t_2, y_2) = (1, 1).$$

We wish to find the parameter c that minimizes

$$f(c) = \frac{1}{2}\sum_{i=1}^{2} \left(y(t_i;c) - y_i\right)^2 .$$

Starting from $c^{(0)} = 1$, perform one iteration of the Gauss–Newton method. Show your work and give the updated parameter $c^{(1)}$.